LINGAP: Legal Intake Navigator for Guidance and Assistance Program

TEAM PLASTICS

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Blog

In partial fulfillment of the requirements in CS 191

CS 191 - WFR1
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Customer Statement of Requirements

The University of the Philippines Office of Legal Aid (OLA) serves as the primary legal outreach arm of the UP College of Law, dedicated to providing free legal representation, counsel, and assistance to indigent litigants who cannot otherwise afford private legal services. Under its traditional operational scheme, OLA operates predominantly on a walk-in-first basis. Applicants seeking legal assistance travel physically to the OLA offices located at Malcolm Hall in UP Diliman or the UP BGC campus. Upon arrival, they queue to undergo an initial manual screening conducted by Law Student Practitioners (LSPs) under the supervision of faculty attorneys. This manual workflow requires LSPs to collect personal background details, administer paper-based indigency evaluations, and evaluate whether the applicant's legal issue falls within the official mandate and operational scope of OLA.

This traditional walk-in intake model creates severe operational bottlenecks and administrative inefficiencies for all participating stakeholders. A critical problem is that zero percent of prospective clients are pre-screened prior to visiting the office. Consequently, many indigent applicants endure long, costly, and time-consuming journeys from distant municipalities only to discover during their initial interview that they are ineligible for OLA's services. Ineligibility often stems from exceeding the strict household income thresholds set for indigency, seeking assistance for matter types outside OLA's scope (such as corporate or commercial disputes), or presenting cases that lack legal merit. For an indigent individual or family with limited financial resources, making an unnecessary trip to the office imposes a heavy financial and personal burden.

At the same time, the reliance on manual intake places an unsustainable burden on Law Student Practitioners and OLA administrative personnel. LSPs spend a disproportionate number of clinical working hours manually filling out paper intake sheets, verifying basic qualification criteria, and interviewing applicants who are ultimately turned away. These repetitive administrative tasks consume valuable time that student practitioners could otherwise spend conducting legal research, drafting court pleadings, preparing trial strategies, and handling active casework for qualified clients. Because initial screenings are conducted in person without prior scheduling, the daily volume of walk-in applicants is unpredictable, leading to crowded waiting areas, extended client wait times, and uneven daily workloads for the clinical staff.

Furthermore, OLA lacks a centralized digital management platform to track incoming matters and organize its internal casework workflow. Case records, client background documents, intake notes, and supervisor assignments are maintained through physical paper files and localized spreadsheets. This fragmented record-keeping framework makes case tracking, document retrieval, and client handovers during semester transitions prone to errors, misplaced documents, and unnecessary delays. Assigning new cases to supervising lawyers is handled manually, which makes it difficult to balance caseloads evenly based on lawyer specializations and existing capacity. Additionally, OLA administrators lack real-time data and analytics to track overall intake metrics, monitor processing durations, evaluate client deflection rates, or measure clinical performance indicators accurately.

To address these systemic challenges, our team proposes the development and deployment of LINGAP (Legal Intake Navigator for Guidance and Assistance Platform). LINGAP is an integrated, bilingual web application and matter management system designed to digitize, streamline, and automate OLA's matter intake and case tracking operations. The primary goal of LINGAP is to shift OLA from a walk-in-first operational structure to a digital-first pre-screening model. By allowing prospective clients to evaluate their eligibility online before traveling to the physical office, LINGAP prevents unnecessary travel costs for ineligible applicants, frees up clinical hours for LSPs, and provides OLA leadership with an automated, centralized platform to manage legal matters efficiently.

The design and operational framework of LINGAP are built around the needs of three distinct primary user groups:

- Legal Applicants and Prospective Clients:These are individuals seeking free legal advice or representation. They interact with LINGAP's public portal or in-office digital kiosks to complete a pre-screening assessment in their preferred language (Filipino or English), upload required verification documents, book consultation appointments, and track the status of their submitted applications.
- Law Student Practitioners (LSPs): LSPs are law students enrolled in the clinical legal education program who conduct client interviews, prepare case evaluations, and handle representation under legal supervision. They use LINGAP's internal dashboard to view pre-screened client details, access uploaded supporting documents prior to interviews, manage

consultation schedules, record interview notes, and track assigned legal matters.

- Supervising Lawyers and OLA Administration: Supervising attorneys and administrative leaders oversee clinical operations, approve case acceptances, and direct legal strategy. They utilize LINGAP to receive automatically triaged case referrals based on legal specializations, monitor active student and lawyer workloads, review performance metrics, and oversee platform security and data privacy compliance.

To enhance intake handling, communication tracking, and user support, LINGAP can incorporate or integrate osTicket, a flexible open-source customer support ticketing platform. By leveraging an osTicket backend structure or mirroring its ticketing architecture, incoming client submissions, general legal inquiries, and document verification requests are converted into distinct, traceable support tickets. Each ticket is automatically assigned a unique control tracking number, allowing LSPs and administrators to route inquiries to specific legal units, track ticket resolution times, attach communication logs, and maintain a complete audit trail of all client interactions. Integrating a ticketing structure ensures that no applicant inquiry is overlooked and provides a clear, systematic workflow for managing client communications from initial contact through case assignment.

The complete end-to-end workflow of the LINGAP system guides users through a structured, multi-stage digital pipeline designed to ensure clarity, convenience, and complete regulatory compliance:

1. Access and Language Selection: Applicants access the LINGAP platform via any web browser on a smartphone, computer, or an in-office kiosk device. Upon landing on the homepage, users select their preferred working language—Filipino or English—ensuring that the platform remains accessible to individuals of varying literacy levels.
2. Data Privacy and Express Consent: Before entering any personal details, applicants are presented with a clear, plain-language Data Privacy Notice in accordance with Republic Act No. 10173 (the Data Privacy Act of 2012). The notice explicitly informs users that OLA collects personal information solely to evaluate legal and financial eligibility, that data is securely stored and accessible only to authorized personnel, and that submitting a form does not create a formal attorney-client relationship. Applicants must review and accept these terms before proceeding.

3. Bilingual Pre-Screening and Indigency Assessment: The applicant completes a multi-step digital form that collects basic personal identification, contact details, geographical location, and representation preferences. The form then executes a dynamic indigency evaluation algorithm that asks for total household monthly income, household size, and real property ownership details. Dynamic skip logic adjusts subsequent questions based on prior answers to streamline entry.
4. Automated Eligibility Evaluation and Deflection: Upon completion, LINGAP's logic engine evaluates the entered data against OLA's predefined mandate criteria and income thresholds. If the applicant is identified as ineligible (e.g., income exceeds indigent standards or the legal issue is out of scope), the system instantly displays feedback explaining the outcome and offers alternative resources or referral details. This automated deflection saves ineligible applicants from making unnecessary trips to Malcolm Hall.
5. Document Upload and Appointment Scheduling: If the applicant passes the preliminary pre-screening, the system prompts them to upload encrypted digital copies or photos of required supporting documents, such as a Barangay Certificate of Indigency or relevant legal notices. The applicant then chooses an available date and time slot from an interactive calendar to schedule an in-person or virtual consultation with OLA.
6. Ticket Generation and Automated Notification: Upon submission, LINGAP generates a unique application reference number (e.g., OLA-2026-0001) and creates an internal tracking ticket. The system automatically sends email notifications containing confirmation details and booking summaries to the applicant, as well as notification alerts to the designated OLA intake team.
7. Automated Triage and Case Assignment: The system's triage engine evaluates the nature of the legal issue (e.g., Family Law, Labor Law, Criminal Law, or Civil Claims) and routes the matter to an active supervising lawyer specializing in that domain who currently holds the lightest caseload. The assigned lawyer and LSP team can then review the pre-screening summary and uploaded documents prior to the scheduled client interview.
8. Application Status Inquiry: Applicants can return to the LINGAP platform at any time to track their application progress. By navigating to the status page and entering their reference number and registered email address, clients can view real-time updates regarding their interview schedule, case review status, or referral outcome.

The development and deployment of LINGAP rely on several foundational tools, reference standards, and open-source systems:

- Data Privacy Act of 2012 (RA 10173): Serves as the primary regulatory framework governing data minimization, user consent, encrypted data transmission, role-based access control, and data retention policies.
- osTicket System Architecture: Provides an open-source model for ticket creation, status tracking, automated email dispatch, and queue management to organize incoming client inquiries and intake submissions efficiently.
- Bilingual Content Framework: Incorporates standardized Filipino and English legal terminology developed in collaboration with legal experts to ensure plain-language clarity for non-lawyer users.
- UP Infrastructure and Web Platforms: Utilizes university hosting environments, security protocols, and academic collaboration frameworks between the UP College of Law and College of Engineering.

To satisfy OLA's operational and technical objectives, LINGAP must meet the following explicit system requirements:

- The system shall present a responsive, bilingual user interface in Filipino and English that allows users to toggle seamlessly between languages at any stage of the pre-screening process.
- The system shall enforce explicit, plain-language data privacy consent screens compliant with RA 10173 prior to capturing any personal user information.
- The system shall implement dynamic skip logic within the pre-screening form to adjust questions based on user input, assessing financial indigency through household income, family size, and asset ownership parameters.
- The system shall instantly generate automated eligibility feedback upon form completion, directing ineligible applicants to alternative legal aid resources or government agencies.
- The system shall provide an interactive online scheduling module that allows eligible applicants to select, book, and reschedule consultation appointments based on OLA's real-time availability.
- The system shall incorporate an encrypted document portal enabling applicants to upload supporting files (such as PDFs or images of indigency certificates and legal notices) securely.

- The system shall automatically generate unique application reference numbers for every submitted pre-screening form to facilitate tracking and record identification.
- The system shall integrate an accessible ticketing system architecture, such as osTicket, to manage incoming client inquiries, convert pre-screenings into manageable tickets, and maintain communication logs.
- The system shall automatically send transactional email notifications to applicants, LSPs, and supervising attorneys to communicate booking confirmations, status changes, and referral assignments.
- The system shall feature an automated triage engine that routes incoming eligible matters to supervising lawyers based on legal specialization and current active caseload limits.
- The system shall offer an offline kiosk mode with simplified data entry screens, allowing OLA staff to assist walk-in applicants who lack personal devices or internet access.
- The system shall feature a public application status lookup tool that enables applicants to query their case status securely using their reference number and email address.
- The system shall provide an administrative dashboard displaying key performance indicators (KPIs), including total submission counts, pre-screening deflection rates, average intake processing durations, and case distribution charts.
- The system shall automatically purge all stored personal data and submitted forms of ineligible applicants after 30 days to adhere strictly to data minimization and privacy standards.

Preliminary UI Design

The LINGAP platform features two primary user-facing portals: a public portal for indigent applicants and an internal administrative dashboard for Law Student Practitioners (LSPs), supervising lawyers, and OLA administrators. The design focuses on high readability, plain-language guidance, accessibility across mobile and desktop devices, and bilingual support in Filipino and English.

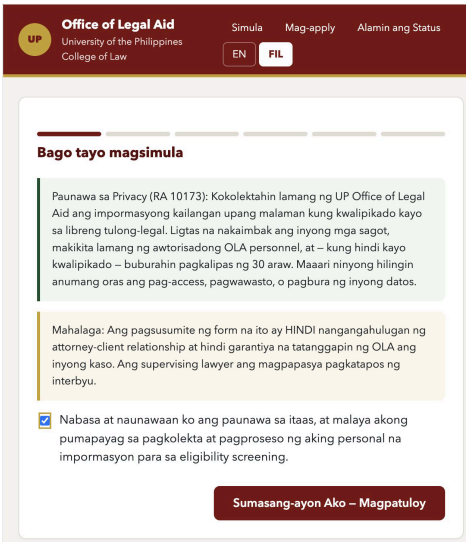
Public Client Portal Navigation Path

Prospective clients access the public portal via a browser or an in-office kiosk to complete pre-screening, submit documents, and check application status.

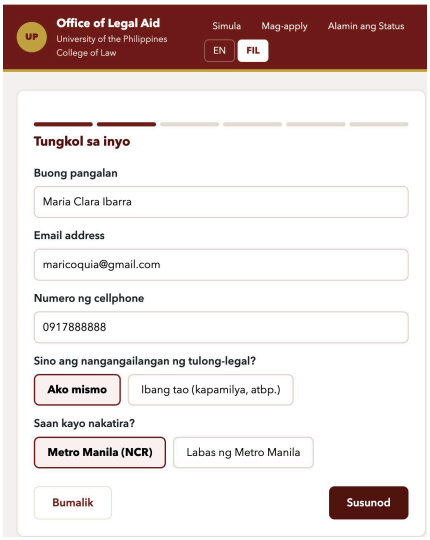
- Landing & Public Homepage: Serving as the main entry point, the page introduces the service, provides a bilingual language switcher (EN/FIL), and presents clear call-to-action buttons (Simulan ang Pre-Screening and Alamin ang Status ng Aplikasyon). It outlines the three-step workflow (Form Completion, Appointment Booking, and Automatic Triage) and provides direct links to the physical office kiosk mode.
    - Parameters / User Inputs: Language toggle (EN or FIL), action selection (Simulan ang Pre-Screening or Alamin ang Status ng Aplikasyon), and kiosk mode toggle.



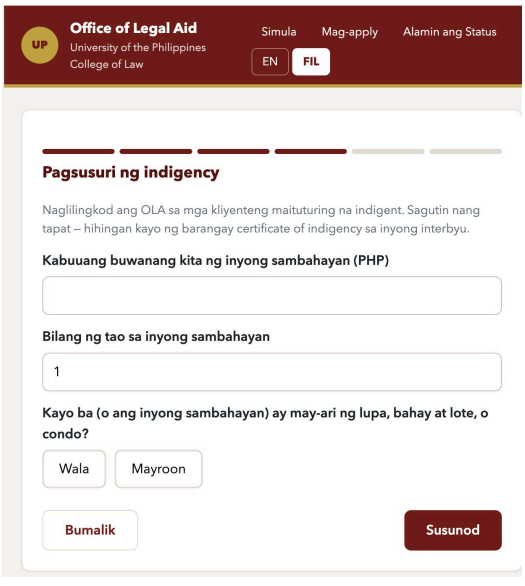
- Privacy & Data Consent Screen: Displays a formal Data Privacy Notice under RA 10173 explaining data retention (30-day deletion for ineligible applicants) and confirming that submission does not establish an attorney-client relationship. Applicants must accept terms to proceed.
    - Parameters / User Inputs: Consent checkbox (Sumasang-ayon Ako - Magpatuloy).



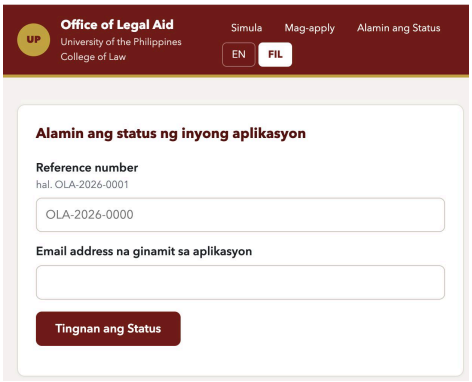
- Applicant Identification Form: Captures personal contact details and geographical parameters to establish basic record identity.
    - Parameters / User Inputs: Full Name (Buong pangalan), Email Address (Email address), Mobile Number (Numero ng cellphone), Beneficiary Context (Ako mismo vs. Ibang tao), and Geographic Location (Metro Manila vs. Labas ng Metro Manila).



- Indigency Screening Assessment Form: Evaluates financial eligibility using dynamic logic to verify if the applicant meets indigent criteria.
    - Parameters / User Inputs: Total Monthly Household Income in PHP (Kabuuang buwanang kita ng inyong sambahayan), Household Member Count (Bilang ng tao sa inyong sambahayan), and Real Property Ownership status (Wala vs. Mayroon for land, house and lot, or condo).



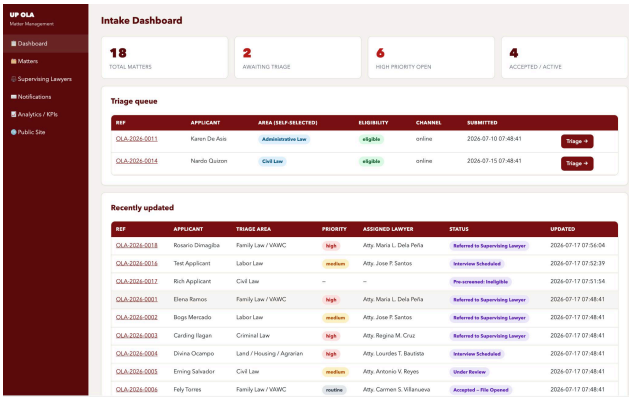
- Application Status Lookup Screen: Allows returning applicants to track case progress independently.
    - Parameters / User Inputs: Reference Number (Reference number, e.g., OLA-2026-0001) and registered email address (Email address na ginamit sa aplikasyon).



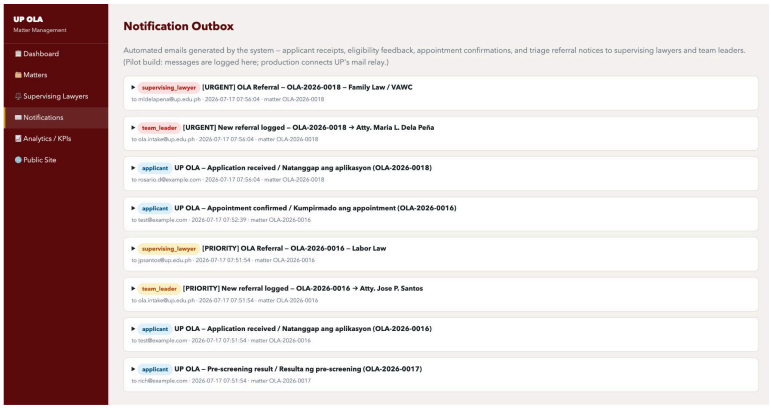
Administrative & Matter Management Portal Navigation Path

Authorized OLA personnel log in to manage intake queues, view automated referrals, send notifications, and analyze clinical metrics

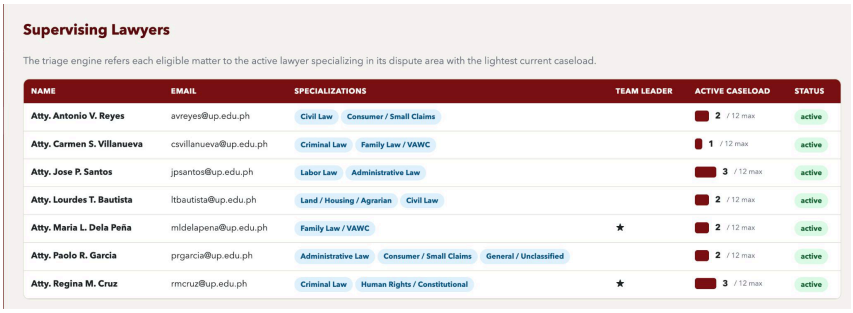
- Intake & Triage Dashboard: Provides an aggregate summary of total matters, active triage queues, and case priority levels. LSPs and lawyers can review incoming submissions, verify self-selected legal dispute areas, and trigger triage actions (Triage →).
    - Parameters / User Inputs: Triage action buttons, priority filters (high, medium, routine), case status updates (Under Review, Interview Scheduled, Referred to Supervising Lawyer), and reference search queries.



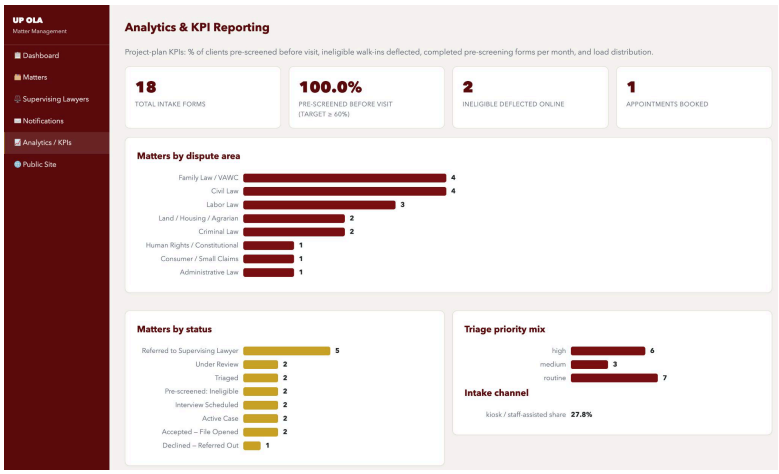
- Notification Outbox Log: Logs all system-generated email communications dispatched to applicants and legal teams, including receipts, appointment updates, and urgent triage referrals.
    - Parameters / User Inputs: Mail dispatch logs, message subject filters, recipient addresses, and mail relay configuration toggles.



- Supervising Lawyer Assignment View: Displays active supervising attorneys, their legal specializations (e.g., Family Law/VAWC, Labor Law, Criminal Law), and current caseload limits (e.g., 2/12 max) to assist automatic or manual case routing.
    - Parameters / User Inputs: Active attorney status toggles, specialization tags, maximum caseload limits, and team leader designations.



- Analytics & KPI Dashboard: Visualizes system performance metrics, such as total forms completed, percentage of clients pre-screened before office visits, ineligible applicant deflection counts, dispute area distribution charts, and intake channel breakdowns (kiosk/staff-assisted vs. online).
    - Parameters / User Inputs: Reporting date range selectors, KPI target benchmarks, and export/filtering options.



Plan of Work

A. Estimate Cost and Resources

1. Human Resources

   The project will be done with a three person software team and two-three person law student practitioners to provide accurate details on the legal side of the software all from UP Diliman. The software team members will be divided into three roles - full stack developer/tech lead, frontend developer, and backend developer. The members should be capable of using SvelteKit, backend APIs, SQL, Git, UI/UX design, authentication, authorization, encryption, and file handling divided between the roles. The legal advice team will provide legal case/document management knowledge like defining legal terminologies and workflow information necessary to make the software as effective and efficient as possible.

2. Reusable Software Resources

   a. Off-the-shelf components

      The ticketing system component can be sourced on open-source platforms such as github. One possible resource for it is this open source ticketing system on github. Aside from that, authentication can be done with Google APIs and integrated with the user's own Google accounts. SMS verification is also possible with existing softwares like Textbee.

   b. Full-experience components

      The software team also has previous experience in developing websites with authentication, forms similar to the project as required in CS 173 Cloud Computing class which can be used in

some components of the projects. The team also has ample experience in AI and data science as required in CS 180 class. The client has also produced a prototype UI design using generative AI which could serve as a starting point for the project.

c. Partial-experience components
   The software team has previous experience in developing appointment booking for veterinarians and pet owners. Although this app was developed like an online medical appointment booking which would require significant modification.
d. New components
   The customizations required by the client as well as legal features according to the Philippine Laws will be created as new components depending on the input of the legal team.

3. Environmental Resources

Resource	**Estimated Specification**	**Purpose**
Personal Computers	8GB RAM, 4-core CPU, 256 GB SSD	Development
Development/Production Server	2-4 vCPU, 4-8GB RAM, 50GB SSD	Shared testing, system deployment
Database Storage	20-30 GB	Database
Document/File Storage (prototype level)	100GB	Document uploads
Backup Storage	200GB	File and system backup
Internet	>= 50 Mbps	Hosting and accessing
Dev Tools - IDE, Git, framework, UI/UX designing tools	Accessible by all members	Dev support

4. Effort Estimation

A. Function Point-Based Estimation

Information Domain Value	Opt	Likely	Pess.	Est. Count	Weight	FP Count
# of external inputs	6	8	11	8	4	32
# of external outputs	4	6	9	6	5	30
# of external inquiries	4	6	9	6	4	24
# internal logical files	4	5	7	5	10	50
# of external interface	0	1	2	1	5	5
Count Total						141

Factor	Value
Backup and recovery	4
Data communications	3
Distributed Processing	1
Performance critical	2
Existing operating environment	4
Online data entry	5
Input transaction over multiple screens	2
Master files updated online	5
Information domain values complex	3
Internal processing complex	3
Code designed for reuse	2
Conversion/installation in design	1
Multiple installations	1
Application designed for change	4
Value adjustment factor	**1.05**

FP = 141 * 1.05 = 148

Based on the team's prior development experience and expected availability, a productivity rate of 9 FP/person-month was assumed, equivalent to approximately 3 FP per developer per month.

Effort = 148/9 = 16.44 person-months

Effort divided into a team of three total 5.48 months, which is a generous estimate for the entire duration of the project (CS 191 - CS 192).

The critical features are estimated to be around 30% of the total FPs which should be around 44 FP. The effort required for the initial release is then computed as:

Effort = 44/9 =4.889 person-months

Thus, the initial software release is estimated to take about 1.63 months for a three person team.

B. Use Case Estimation

Subsystem	Use Cases	Scenarios	Pages	Estimated LOC
User & Authentication	4	6	5	800
Case Management	6	10	8	1800
Document Management	4	7	6	1500
Analytics & Reporting	3	5	4	1200
Security & Administration	4	10	8	900
Total				6200

The estimated LOC is significantly lower compared to the prescribed estimation table in Pressman, as the use of Svelte +

Tailwind/Typescript + SQL would reduce the code needed to be written.

The productivity rate based on LOC via converting the productivity rate of 9FP/person-month is assumed to be 400 LOC/person-month. The effort is computed to be

Effort = 6200/400 = 15.5 person-months.

The effort is also divided into three persons, the project is estimated to take 5.17 months. The critical features will take 1.55 months.

C. Reconciliation

Since the effort estimated using the FP-based estimation and use-case estimation resulted in approximately similar results, it suffices to say that the values are valid estimates. Additionally, as these are just estimates, the actual effort needed for the project may not be accurately represented.

5. Cost Estimation

The project heavily relies on open-source, free, and/or existing systems. Additionally, the team already has experience production using freewares which should not incur additional production costs. However, the actual deployment of the software might require deployment and maintenance cost. Nevertheless, as the project will be an internal project of the university, the deployment of the software will be subsidized by the university itself.

B. Identify Schedules and Milestones

The full GANTT chart can be found through this link: Project GANTT Chart.

C. Project Metrics

Task completion – check how many planned tasks are finished.
Time spent – compare the actual time spent on tasks with the estimated time.

Errors found – keep track of errors found during development and testing.
Defects – track problems that are still present in the system.
Schedule progress – check if we are still following the project schedule.

D. Effort Breakdown

Team Member	**Modules / Classes**	**Responsibilities**
Buton, Kylian	**User/Application Module** — User, Applicant, Application, ReferenceNumber	Applicant registration, application submission, reference number generation, and viewing application information.
Pulbosa, Mariel Sofia	**Scheduling & Consultation Module** — Schedule, Appointment, Consultation, Availability	Consultation scheduling, available time slots, appointment creation, and rescheduling/cancellation.
Rebollido, Pauline Mae	**Case Management & Triage Module** — Case, CaseAssignment, Lawyer, Triage	Case creation, case status tracking, automated case assignment, and lawyer/case management.

Integration Coordinator: Pulbosa, Mariel Sofia will coordinate the integration of all modules and make sure the modules work together properly.

Integrated System Testing: Buton, Kylian and Rebollido, Pauline Mae will perform the testing of the integrated system together

Project Risks

Risk	Category	Probability	Impact
Project requirements may change during development	PS – Project Size/Scope	60%	2 – Critical
The three-person development team may not have enough time to complete all required features	PS – Project Size/Staffing	60%	2 – Critical
Legal requirements and OLA workflows may not be fully understood by the software team	CU – Customer/User	40%	2 – Critical
Law student practitioners may have limited availability for consultations and feedback	CU – Customer/User	60%	2 – Critical
The automated indigency and eligibility rules may be incorrectly implemented	TE – Technology	40%	1 – Catastrophic
The automated case triage system may incorrectly assign matters to supervising lawyers	TE – Technology	35%	2 – Critical
Secure handling of sensitive applicant information and uploaded legal documents may be difficult to implement correctly	TE – Technology	40%	1 – Catastrophic
Integration with osTicket, email services, or other external software may not work as expected	TE – Technology	50%	2 – Critical
Appointment scheduling system may conflict with OLA's actual availability or existing scheduling process	PS – Project/Process	40%	2 – Critical
Bilingual Filipino/English interface may contain unclear or inaccurate legal terminology	CU – Customer/User	40%	2 – Critical
Offline kiosk mode may require more development effort than initially estimated	PS – Project Size	50%	3 – Marginal

Team members may have insufficient experience with some required technologies or security requirements	ST – Staff	40%	2 – Critical
A team member may become unavailable during development, increasing the workload of the remaining members	ST – Staff	30%	2 – Critical
Network infrastructure (UP) may experience availability or configuration problems	DE – Development Environment	30%	2 – Critical
Testing may take longer than expected because the system contains many interconnected modules	PS – Project Size	60%	2 – Critical
Project deadline may be tightened or development time may be reduced	PS – Schedule	40%	2 – Critical